Ömer Faruk İşeri

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Education

Istanbul University-Cerrahpasa | Istanbul, Turkey *(2020-2025)*

B.Sc. in Computer Engineering, *Graduated: June 2025*

Experience

GoLive **| İstanbul, Turkey** *(August-September 2024)*

Intern (Full-time)

- Managed a complex RAG system lifecycle, demonstrating proficiency in Python, robust CI/CD, and scalable deployment (Docker/AWS EC2).

Carbon Consulting **| İstanbul, Turkey** *(December 2024-March 2025)*

Intern (Part-time)

- Worked on SAP-integrated HR chatbot for Microsoft Teams with Selenium testing and LLM prompt optimization (70.3%→94.5% accuracy).

Sollen Technologies **| İstanbul, Turkey** *(November 2025-Present)*

Fullstack Developer

- Developing a full-stack B2B Occupational Health and Safety (OHS) platform using Python (FastAPI), React, and Flutter, focusing on scalable backend architecture and real-time reporting to streamline enterprise safety compliance.

Projects

Tile Tactics: 2048 Card Puzzle – Mobile Game *(Released on Google Play & Itch.io)*| 3 Months | Solo Developer

- Designed and developed a unique hybrid puzzle game, blending core 2048 mechanics with a deep roguelike card and resource management system (average session time: 10-12 minutes).
- Architected modular codebase using Manager Pattern (8 specialized managers), State Pattern for centralized game state, and Observer Pattern for event-driven systems.
- Implemented weighted random card distribution system with Strategy Pattern, adapting card pools based on game phase (Early/Mid/Late) for balanced progression.
- Managed game data and content using ScriptableObjects for rapid iteration and decoupled component design, streamlining the balancing pipeline.
- Designed and balanced 27 unique cards with mana economy, 14 stackable perks with conflict resolution logic, and dynamic multiplier chain system for exponential scoring and satisfaction.
- Added an achievement system that rewards players with gold for completing milestones.
- Integrated Unity Ads (LevelPlay SDK) for rewarded video monetization with ad-based continue mechanic.
- Acquired 180+ active users with an average rating of 4.5/5 stars.

Neon Blast – Arcade Roguelike Shooter *(Released on Google Play & Itch.io)* | Solo Developer

- Designed and engineered a high-performance, ultra-lightweight (<3MB) custom 2D rendering engine in Vanilla JavaScript/HTML5 Canvas, achieving stable 60 FPS on mobile devices.
- Architected a memory-efficient game loop using Object Pooling and Manager Pattern (BossManager, SpawnManager) to handle 200+ active entities simultaneously without garbage collection spikes.
- Optimized collision detection using Grid-Based Spatial Partitioning, reducing computational complexity from $O(N^2)$ to $O(N)$ to support 200+ active entities and 500+ projectiles.
- Designed 7 unique multi-phase boss fights using Polymorphic class structures and Finite State Machines for complex enemy AI behavior.
- Built a custom localization system supporting 5 languages with an automatic locale detection.
- Integrated Firebase Firestore for a leaderboard system (Monthly, Weekly, and All-time).
- Implemented a custom state management system to handle input, physics, and rendering pipelines, minimizing overhead without external frameworks.
- Implemented a dynamic difficulty scaling algorithm that linearly interpolates enemy spawn rates (from 1000ms down to 200ms) and adjusts probability weights for 8 distinct enemy classes based on a cumulative difficulty multiplier.

HELLRAZE – Roguelike FPS Game *(Released on Itch.io)* | Solo Developer

- This project was awarded **"Best Technical Implementation"** at the **2025 Game Development Project Days**, organized by the Department of Computer Engineering at Istanbul University-Cerrahpaşa.
- Developed procedurally generated boomer shooter with varied dungeon layouts, enemy AI, and boss fights using state machines.
- Implemented 7 weapon systems (shotgun, flamethrower, grappling hook, etc.) with unique mechanics.
- Designed and implemented enemy AI utilizing State Machines for complex behaviors, including pathfinding, threat assessment, and varied attack patterns.
- Created a meta-progression system and in-game upgrade economy to enhance long-term replay
- Created 20+ scalable perks to ensure high gameplay variety across runs.
- Designed and implemented a **custom occlusion culling system in Unity** to optimize performance for procedurally generated room rendering.
- Documented the development process through devlogs on YouTube, engaging with the indie game community.

Educational Game for Children with Special Needs *(Released on Itch.io)* | Group Project

- Developed four core gameplay modes (matching, object selection, sequencing daily routines) across three difficulty levels and nine thematic categories.
- Engineered a ScriptableObject-based modular system for scalable content creation and optimized asset management.
- Built a teacher dashboard with analytics, progress tracking, and visual behavior reports using persistent JSON data.
- Designed UI/UX for accessibility, focusing on color contrast, intuitive touch controls, and auditory feedback.

Activities

IEEE (Institute of Electrical and Electronics Engineers) Society, Computer Science Subbranch
(September 2022 - January 2023)
Active Member

- Participated in game development workshops

Skills

Game Engines & Tools: Unity (Advanced – 2D & 3D), Plastic SCM, (Unreal Engine 4/5 – Beginner)
Programming Languages: C#, Python, C++
Gameplay & Systems Development: Gameplay Systems, Weapon & Combat Systems, Enemy AI (Finite State Machines, Pathfinding), Progression & Meta-Progression Systems, Game Economy & Combat Balancing, Procedural Generation, Level Design, Game UI/UX
Performance & Optimization: Object Pooling, Memory Profiling, Occlusion Culling, Grid-Based Spatial Partitioning, Garbage Collection Optimization
Architecture & Patterns: OOP, Design Patterns (State, Observer, Strategy, Manager, Factory), ScriptableObjects, Coroutines, Event-Driven Systems
Production & Workflow: Git, Agile / Scrum, Technical Debugging, Profiling & Optimization
Languages: Turkish (Native), English (Advanced), French (Beginner)